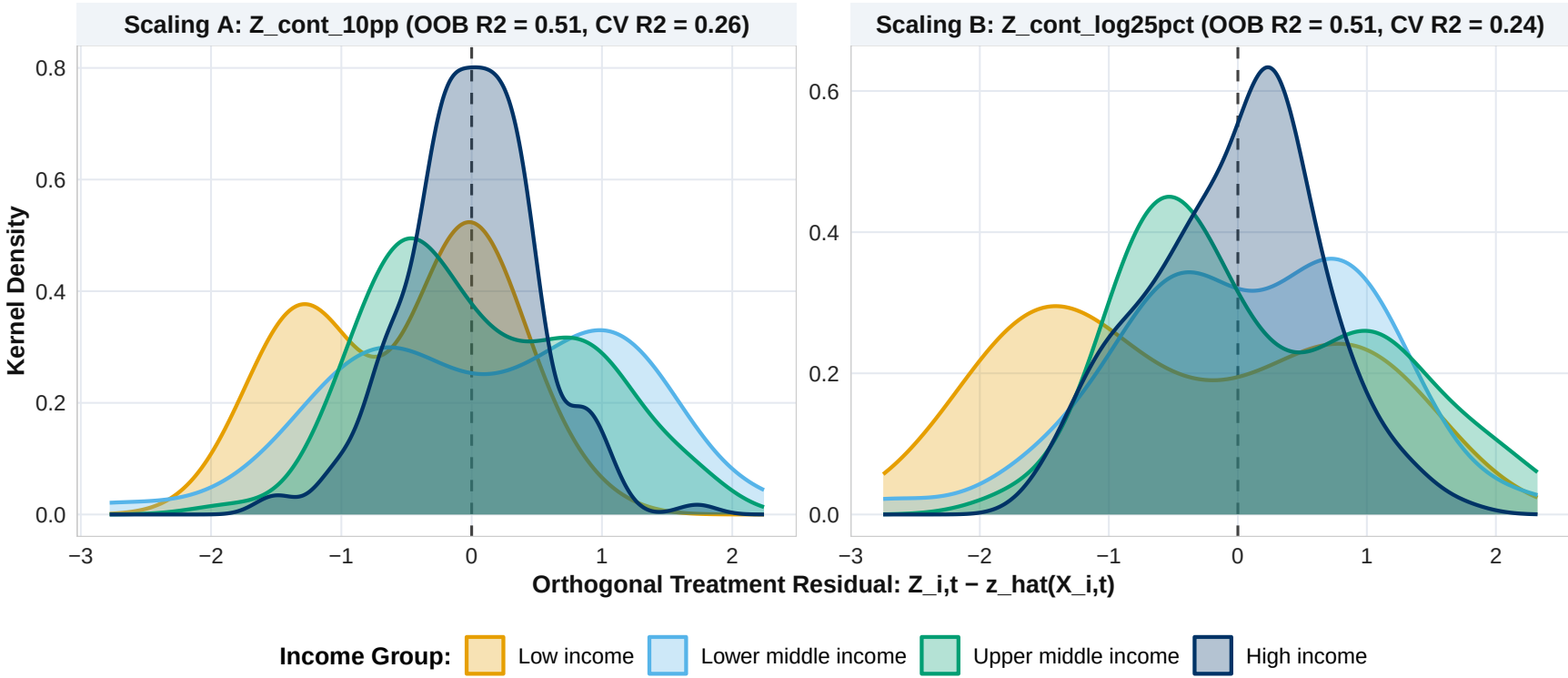
Orthogonal Treatment Residual Overlap (Z_i,t − z_hat(X_i,t)) Across Income Tiers
Centred residual variation available for causal identification after Random Forest conditioning
Scaling A: Z_cont_10pp (OOB R2 = 0.51, CV R2 = 0.26)
Scaling B: Z_cont_log25pct (OOB R2 = 0.51, CV R2 = 0.24)
Kernel Density
0.0
0.2
0.4
0.6
0.8
−3
−2
−1
0
1
2
Orthogonal Treatment Residual: Z_i,t − z_hat(X_i,t)
Income Group:
Low income
Lower middle income
Upper middle income
High income